

CAPE REINGA, New Zealand

WMO: 930040

Lat: 34.4294S

Lon: 172.6817E

Elev: 213

StdP: 98.80

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	9.3	3.0	4.8	10.0	4.2	5.2	10.4	24.4	13.0	22.4	12.2	8.6	200	0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.7	23.2	19.6	22.5	19.2	21.8	18.8	21.1	22.1	20.4	21.4	19.9	20.8	7.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.9	21.4	20.1	15.2	20.9	19.5	14.6	20.4	62.3	22.0	59.9	21.4	57.9	20.9	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	19.1	16.5	DB	7.0	25.0	0.9	0.8	6.4	25.6	5.9	26.1	5.4	26.5	4.7	27.1	(4)
(5)			WB	4.8	22.3	0.8	0.6	4.2	22.7	3.7	23.1	3.2	23.4	2.6	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	18.8	19.6	18.8	17.3	15.4	13.5	12.7	12.5	13.3	14.0	15.2	17.2	(6)	
	DBStd	2.80	1.31	1.29	1.30	1.39	1.56	1.49	1.36	1.26	1.32	1.26	1.33	1.36	(7)	
	HDD10.0	2	0	0	0	0	0	1	1	1	0	0	0	0	(8)	
	HDD18.3	1069	11	3	11	37	91	144	176	181	152	134	93	39	(9)	
	CDD10.0	2071	273	268	272	218	168	106	83	79	98	124	157	224	(10)	
	CDD18.3	98	25	38	24	5	1	0	0	0	0	0	0	5	(11)	
	CDH23.3	22	5	14	3	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	8.4	8.0	7.6	7.9	7.9	8.8	9.1	9.2	8.8	8.6	8.8	8.3	7.9	(14)	
(15) Precipitation	PrecAvg	1197	58	85	72	95	118	132	148	125	98	80	68	93	(15)	
	PrecMax	1859	166	282	183	308	220	357	317	215	206	155	151	212	(16)	
	PrecMin	627	12	18	9	15	30	56	46	58	51	21	20	13	(17)	
	PrecStd	295	39	71	50	68	49	58	67	45	45	38	34	55	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	24.8	23.6	21.9	20.1	17.7	16.8	16.5	17.7	19.1	20.3	22.4	(19)	
		MCWB	20.0	21.0	19.6	18.5	17.5	16.3	15.1	14.0	14.9	15.9	16.9	19.0	(20)	
	2%	DB	22.8	23.6	22.6	20.9	19.0	16.8	15.9	15.7	16.7	17.9	19.2	21.3	(21)	
		MCWB	19.3	20.2	19.1	18.2	17.1	15.7	14.4	13.8	14.3	15.1	16.4	18.1	(22)	
	5%	DB	22.1	22.8	21.9	20.1	18.2	16.2	15.2	15.0	16.0	16.9	18.4	20.4	(23)	
		MCWB	18.9	19.5	18.7	17.9	16.6	15.0	13.7	13.3	14.0	14.6	15.9	17.7	(24)	
	10%	DB	21.3	22.0	21.1	19.4	17.6	15.6	14.6	14.5	15.3	16.2	17.6	19.6	(25)	
		MCWB	18.5	19.1	18.4	17.4	16.1	14.4	13.2	12.9	13.7	14.3	15.4	17.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	22.2	21.3	20.5	18.8	17.2	16.1	15.6	16.1	16.8	18.5	20.2	(27)	
		MCDB	22.4	23.2	22.1	20.8	19.1	17.3	16.3	15.8	16.7	18.0	19.1	21.0	(28)	
	2%	WB	20.6	21.5	20.4	19.6	17.9	16.3	15.1	14.7	15.2	16.1	17.4	19.3	(29)	
		MCDB	21.7	22.5	21.3	20.0	18.4	16.5	15.4	15.0	15.8	16.8	18.1	20.0	(30)	
	5%	WB	19.9	20.8	19.8	18.9	17.3	15.6	14.4	14.1	14.8	15.5	16.8	18.7	(31)	
		MCDB	21.0	21.8	20.8	19.4	17.8	15.9	14.8	14.5	15.4	16.3	17.5	19.4	(32)	
	10%	WB	19.3	20.1	19.1	18.1	16.6	14.9	13.7	13.4	14.2	15.0	16.2	18.2	(33)	
		MCDB	20.5	21.0	20.4	18.8	17.3	15.4	14.3	14.1	15.0	15.8	17.1	18.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	4.7	4.6	3.9	3.4	3.1	3.2	3.5	3.9	4.2	4.4	4.5	(35)	
		MCDBR	5.5	5.7	5.5	4.7	3.9	3.3	3.6	4.0	4.5	5.1	5.2	5.4	(36)	
		MCWBR	3.2	3.2	3.2	3.2	2.9	3.1	3.1	3.1	3.3	3.5	3.5	3.5	(37)	
	5% WB	MCDBR	4.4	4.3	4.3	3.4	3.2	2.7	2.9	3.1	3.6	4.1	4.2	4.0	(38)	
		MCWBR	3.0	3.0	3.0	2.8	2.8	3.1	3.1	2.9	3.0	3.3	3.4	3.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.336	0.335	0.323	0.313	0.304	0.304	0.300	0.308	0.326	0.326	0.326	0.332	(40)	
	taud		2.621	2.654	2.679	2.699	2.693	2.662	2.680	2.647	2.572	2.587	2.614	2.647	(41)	
	Ebn at Noon		1001	982	958	915	871	841	866	906	938	980	1004	1008	(42)	
	Edh at Noon		101	94	86	75	66	64	66	77	93	99	101	99	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.11	6.35	5.17	3.83	2.76	2.27	2.45	3.33	4.53	5.80	6.60	6.84	(44)	
	RadStd		0.41	0.46	0.28	0.23	0.13	0.13	0.14	0.17	0.24	0.32	0.38	0.54	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page